Advanced Hardware Engineering

Seminar: Emerging Trends in IP Design and FPGAs

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Trends in modern VLSI Design

- AI/ML Integration
  - AI assists in design automation, synthesis, and verification
  - Data-driven optimization approaches
  - Circuit foundation models emerging
  - Benefits: generalization, faster design cycles
- Design Tool Evolution
  - AI/ML integrated into EDA tools
  - Cloud-based collaborative workflows
  - Advanced automation for timing closure, synthesis
  - Scalable solutions for advanced nodes

Trends in modern VLSI Design

- Security-First Hardware
  - Hardware-level security measures
  - Secure boot and hardware root-of-trust
  - Side-channel resistance, IP protection
  - Security integrated early in design flows
- Heterogeneous Integration & 3D ICs
  - Chiplet-based and 3D stacking architectures
  - Increased performance and modularity
  - IP reuse and scalability
  - Thermal and testing challenges

Trends in modern VLSI Design

- Manufacturability & Yield
  - Process variation at advanced nodes
  - Design for Manufacturing (DFM) and Test (DFT)
  - Variation-aware design practices
  - Yield optimization techniques
- Verification and Complexity
  - Verification dominates design cycles (80% effort)
  - Timing closure challenges: routing, sizing
  - Signal integrity, cross-talk issues
  - Formal verification and assertion-based methods

Trends in modern VLSI Design

- Skills & Cross-Disciplinary Demands
  - Knowledge across digital, analog, layout, verification
  - Programming and AI/ML familiarity
  - Communication and collaboration
  - Lifelong learning essential
- Power and Energy Opimization
  - Power constraints central to design
  - Techniques: multi-Vt, power gating, DVFS
  - Focus on mobile, IoT, and sustainability
  - Balancing performance and power budget

Seminar Work

- Each Team will got a topic
  - Describe the selected topic
  - Research
  - Analysis
- Setup a Github repository
- Everyone will write a research paper about his topic (my recommendation use LaTeX))

Seminar Work

- Use latex and bibtex
- See ieee template at: https://www.ieee.org/conferences/publishing/templates.html
- 6-8 pages
- Apply the methods and techniques from scientific writing
- (Use citavi or Zotero for managing your references)
- Write for each related reference a short note of how it is related to your seminar topic – the note is part of your bibtex management tool and/or an extra bibtex entry.

Timeline and milestones

Milestone	Content	Date
1 Team and topic	1 Selection per Team (only one student selects for the team)	29.10.2026
2 Outline	Research question, structure, at least 10 sources	05.11.2026
3 First draft	First version incl. Main figures and references	19.11.2026
4 Peer review and polishing	Teams review each other with a checklist	03.12.2026
5 Final paper	6-8 pages in IEEE format as PDF	31.12.2026
6 Presentation	20 minutes plus discussion	07.01.2027

Structure of a 6-page IEEE paper

Section	Pages	Content
Abstract	0.2	Problem, approach, results, conclusion in 150 to 200 words
Introduction	0.8	Motivation, problem, contributions, outline
Background and related work	1.0	Concepts the reader needs; state of the art
Main part	2.0	Your analysis, comparison or design
Discussion / evaluation	0.9	Results, limits, lessons for IP design
Conclusion	0.3	Findings and outlook
References	0.8	At least 15 sources in IEEE style

Structure of a 8-page IEEE paper

Section	Pages	Content
Abstract	0.2	Problem, approach, results, conclusion in 150 to 200 words
Introduction	1.0	Motivation, problem, contributions, outline
Background and related work	1.5	Concepts the reader needs; state of the art
Main part	3.0	Your analysis, comparison or design
Discussion / evaluation	1.0	Results, limits, lessons for IP design
Conclusion	0.5	Findings and outlook
References	0.8	At least 15 sources in IEEE style

What makes a good research paper

Clear question

State what the paper answers and why it matters, in the introduction.

Evidence

Support each claim with a reference, data or a clear argument.

Critical analysis

Compare alternatives, show trade-offs and limits.

Logical structure

Every section and paragraph has one purpose.

Multiple authors contributing to a single paper, rather than each author writing a separate part of a paper!

Abstract, introduction and conclusion

Part	Should contain	Avoid
Abstract	Context, problem, approach, findings, conclusion	Citations, undefined abbreviations
Introduction	Motivation, problem, contributions, outline	Long background, vague goals
Conclusion	Findings, limits, outlook	New facts, copied sentences

Two pages of FPGA background in every paper? We will have more pages explaining FPGAs than actually using them!

Literature research and citations

Topic	Guidance
Where to search	IEEE Xplore, ACM DL, Google Scholar, arXiv, conference proceedings
Vendor documents	Good for facts about products; not the only sources
Method	Start with surveys; follow references backwards and forwards
Citation style	Numbered IEEE style, e.g. [1]; one BibTeX entry per source
Example entry	[1] A. Author and B. Author, "Title," in Proc. Conf., City, 2025, pp. 1-6.

I found it on Google” is not a valid scientific reference!

Figures, tables and writing style

Figures and tables

Numbered, with captions, referenced in the text; readable fonts; units on axes.

Writing style

Short sentences, active voice, abbreviations defined, consistent terms.

Checks

Spell-check, read aloud, ask a reader outside the team.

Just because a figure is AI generated doesn't mean the reader understands it!

Teamwork and tools

1. **Set up the project** – IEEE template in Overleaf or Git, shared BibTeX file (dont forget the invitation)
2. **Split the work** – research, analysis, figures, editing; everyone writes
3. **Meet regularly** – short meetings with a task list and deadlines
4. **Merge and harmonise** – one editor checks style, terms and references
5. **Review** – internal review before each milestone

3-4 Students: One paper!

Rules: originality and AI tools

Originality

Write in your own words; cite every source; mark quotes.

AI tools

Declare the use, verify all facts and references yourself.

Folloe the university guidelines!

Figures

Draw your own or cite the source; observe copyright.

Evaluation criteria

70% Paper – 30% Presentation

For the report:

Criterion	Weight	What is assessed
Content and depth	30 %	Topic understood; expected points covered
Critical analysis	20 %	Comparisons, trade-offs, limits
Structure and clarity	20 %	Logical flow; abstract, introduction, conclusion
Language and figures	15 %	Style, correctness, quality of figures
References	10 %	Quality, number, IEEE format
Teamwork and process	5 %	Milestones, contribution notes

Topic Overview

- IP and System Design with FPGAs
- AI on FPGAs
- Reconfigurability
- Design Methodolgy
- FPGA Applications and Systems

How to choose a topic

1. **Form a team** – 3 to 4 students
2. **Read the topic slides** – expectations, keywords and difficulty
3. **Check the literature** – at least five relevant papers should exist
4. **Submit your selection** – Starts on 26.10.2026 – 5 p.m. to 29.10.2026 – 09:00 a.m.
5. **Receive the assignment** – each topic is given to one team

From DPU to NPU: AMD's AI Inference IP

Topic 1 · AI on FPGAs

AMD's Vitis AI moved from the Deep Learning Processor Unit (DPU) in programmable logic to a Neural Processing Unit (NPU).

Expected in your paper:

- Explain the DPU architecture, its data flow and the toolchain (quantizer, compiler, runtime)
- Describe the NPU, what changed and which devices are supported
- Compare capabilities, supported models, flexibility and limits
- Discuss consequences for designers on smaller devices

Starting points: Vitis AI, DPU, NPU, VART, Versal AI Edge

Quantization and Low-Precision Neural Networks

Topic 2 · AI on FPGAs

Reducing the bit width of weights and activations is the main lever for efficient inference on FPGAs.

Expected in your paper:

- Compare post-training quantization and quantization-aware training
- Show how bit width (INT8, INT4, binary) affects accuracy, LUTs, DSPs and power
- Evaluate mixed-precision strategies
- Support the analysis with published results or an own experiment

Starting points: Brevitas, QAT, INT8, binarized networks, DSP packing

Open-Source ML Flows: hls4ml and FINN

Topic 3 · AI on FPGAs

Open-source tools turn trained networks into FPGA hardware, using HLS or streaming dataflow architectures.

Expected in your paper:

- Describe both flows from trained model to hardware
- Compare their architecture and the networks they support
- Discuss latency, resource use, supported devices and limits
- Compare with a vendor flow such as Vitis AI

Starting points: hls4ml, FINN, Brevitas, QONNX, dataflow

Transformer and LLM Inference on FPGAs

Topic 4 · AI on FPGAs

Attention-based models are memory-bound and challenge FPGA accelerator design.

Expected in your paper:

- Explain compute and memory demands of attention and the KV cache
- Survey FPGA accelerator architectures and the role of HBM
- Compare latency and energy per token with GPUs using published results
- Discuss feasibility on edge and data-center devices

Starting points: attention, KV cache, HBM, sparsity, energy per token

Spiking Neural Networks on FPGAs

Topic 5 · AI on FPGAs

Event-driven neural networks promise very low power; FPGAs are a flexible platform for them.

Expected in your paper:

- Explain spiking neuron models and event-driven computing
- Show how neurons and synapses map to FPGA resources
- Compare FPGA implementations with neuromorphic chips
- Discuss use cases and limits such as training and accuracy

Starting points: LIF neuron, event-driven, neuromorphic, SNN accelerator

Versal AI Engines and Heterogeneous Programming

Topic 6 · AI on FPGAs

Versal devices combine processors, programmable logic, AI Engines and a network-on-chip.

Expected in your paper:

- Describe the Versal architecture and its components
- Explain the AI Engine programming flow: graphs, kernels, data movement
- Evaluate which workloads benefit; compare with a pure PL design
- Discuss tool complexity and portability

Starting points: Versal, AIE-ML, graph programming, NoC, Vitis

TinyML and Edge AI on Small FPGAs

Topic 7 · AI on FPGAs

Always-on edge AI needs low power, low latency and small size.

Expected in your paper:

- Define requirements for edge AI applications
- Survey small FPGA families and their AI flows
- Compare with microcontroller-based TinyML
- Present an example application with estimated resources

Starting points: TinyML, low-power FPGA, keyword spotting, sensAI

Dynamic Function eXchange: Flow and Use Cases

Topic 8 · Reconfigurability

DFX allows parts of an FPGA design to be swapped at runtime.

Expected in your paper:

- Explain partitions, Pblocks, partial bitstreams and decoupling
- Describe the Vivado flow and its restrictions
- Present use cases and quantify area, power and reconfiguration time
- Discuss pitfalls in floorplanning, timing and debugging

Starting points: DFX, partial reconfiguration, ICAP, PCAP, Pblock

Runtime Management for Reconfigurable Hardware

Topic 9 · Reconfigurability

Reconfigurable modules need loading, scheduling and isolation at runtime.

Expected in your paper:

- Explain why runtime management is needed
- Survey approaches: Linux FPGA manager, device tree overlays, research systems
- Compare features: scheduling, isolation, abstraction
- Derive requirements for an embedded use case

Starting points: FPGA manager, device tree overlay, hardware threads, ReconOS

Fault Tolerance with Partial Reconfiguration

Topic 10 · Reconfigurability

SRAM-based FPGAs are sensitive to single event upsets; reconfiguration can repair them.

Expected in your paper:

- Describe radiation and aging effects on SRAM-based FPGAs
- Explain mitigation: scrubbing, TMR, readback and ECC
- Show how partial reconfiguration enables repair at runtime
- Compare overhead in area, power and recovery time

Starting points: SEU, scrubbing, TMR, soft error mitigation

FPGA Virtualization and Multi-Tenancy

Topic 11 · Reconfigurability

Sharing FPGAs between users requires isolation, scheduling and security.

Expected in your paper:

- Explain virtualization goals: sharing, isolation, scheduling
- Survey approaches: reconfigurable slots, shells, hypervisor support
- Discuss security risks such as side channels and malicious tenants
- Compare cloud implementations and research prototypes

Starting points: shell, role, multi-tenant FPGA, isolation, side channel

Overlays and Coarse-Grained Reconfigurable Arrays

Topic 12 · Reconfigurability

Overlays and CGRAs trade peak efficiency for faster compilation and easier programming.

Expected in your paper:

- Define overlays and CGRAs and their motivation
- Compare with fine-grained FPGA design in productivity and efficiency
- Survey examples and their tool flows
- Discuss when an overlay is better than direct RTL design

Starting points: overlay, CGRA, soft processor array, compile time

State of the Art in High-Level Synthesis

Topic 13 · Design methodology

HLS tools compile C/C++ into hardware, and new compiler infrastructures are changing the field.

Expected in your paper:

- Compare HLS tools in language support and optimizations
- Explain key transformations: pipelining, unrolling, dataflow
- Evaluate quality of results from the literature or a small benchmark
- Discuss the role of MLIR and CIRCT in new flows

Starting points: Vitis HLS, Bambu, MLIR, CIRCT, Dynamatic

Open-Source FPGA Toolchains

Topic 14 · Design methodology

Open-source synthesis and place-and-route tools are maturing for several FPGA families.

Expected in your paper:

- Describe the flow: Yosys, nextpnr and bitstream tools
- Survey supported devices, including the state for Xilinx 7 series
- Compare quality, speed and features with vendor tools
- Discuss benefits for research, teaching and security

Starting points: Yosys, nextpnr, F4PGA, Project X-Ray

Modern Hardware Description Languages

Topic 15 · Design methodology

Hardware construction languages and generators raise productivity compared with VHDL and Verilog.

Expected in your paper:

- Compare Chisel, SpinalHDL, Clash and Amaranth with VHDL
- Explain concepts: embedded DSLs, parameterization, type systems
- Show examples of productivity and verification support
- Discuss adoption barriers in industry

Starting points: Chisel, SpinalHDL, Amaranth, Clash, FIRRTL

Large Language Models for Hardware Design

Topic 16 · Design methodology

LLMs are being used to generate, verify and debug RTL.

Expected in your paper:

- Survey LLM-based RTL generation, verification and debugging
- Explain benchmarks and metrics such as functional correctness
- Analyze limitations: hallucination, synthesizability, verification
- Discuss risks and guidelines for use in IP design

Starting points: LLM, RTL generation, VerilogEval, hallucination

CI and Test Automation for FPGA IP

Topic 17 · Design methodology

Software-style continuous integration is entering hardware development.

Expected in your paper:

- Describe CI/CD pipelines for HDL projects: Git, Tcl, containers
- Compare verification frameworks such as cocotb, VUnit and UVM
- Explain automated regression, coverage and reporting
- Evaluate effort and benefit with a small example

Starting points: CI/CD, cocotb, VUnit, Tcl scripting

Formal Verification of FPGA IP

Topic 18 · Design methodology

Formal methods prove properties of IP that simulation cannot cover completely.

Expected in your paper:

- Explain model checking, assertions and equivalence checking
- Show how properties for bus interfaces such as AXI are written
- Compare with simulation in coverage and effort
- Survey open-source and commercial tools

Starting points: SymbiYosys, SVA, AXI protocol checker, model checking

On-Chip Interfaces Beyond AXI4

Topic 19 · IP and system design

AXI4 is the de facto standard, but coherent and newer interfaces are emerging.

Expected in your paper:

- Compare AXI4, AXI4-Stream, AXI5 and other standards
- Explain channels, handshakes, ordering and coherence
- Discuss interoperability and IP reuse across vendors
- Show a case in which interface choice affects performance

Starting points: AXI5, CHI, Wishbone, TileLink, coherence

Networks-on-Chip in FPGAs

Topic 20 · IP and system design

Buses do not scale to large systems; networks-on-chip are used inside FPGAs.

Expected in your paper:

- Explain the NoC concept and why buses do not scale
- Describe hard NoCs (Versal) and soft NoCs on FPGAs
- Compare latency, bandwidth and area
- Discuss the effect on IP interfaces and system design

Starting points: NoC, Versal NoC, soft NoC, routers

RISC-V Soft Cores and Custom Instructions

Topic 21 · IP and system design

RISC-V offers open soft processors that can be extended with custom instructions.

Expected in your paper:

- Survey RISC-V soft cores and compare them with MicroBlaze
- Explain custom instruction and coprocessor interfaces
- Evaluate area, maximum frequency and performance of a chosen core
- Discuss the ecosystem: toolchains and Linux support

Starting points: RISC-V, VexRiscv, CVA6, custom instruction, MicroBlaze V

IP-XACT and Register Map Generators

Topic 22 · IP and system design

Metadata standards and generators automate IP documentation, RTL and drivers.

Expected in your paper:

- Explain IP-XACT and its role in the Vivado IP packager
- Survey register-map description tools such as SystemRDL
- Show how automation reduces errors in documentation, RTL and drivers
- Propose a workflow for an IP team

Starting points: IP-XACT, SystemRDL, register map, driver generation

Chiplets, HBM and Multi-Die FPGAs

Topic 23 · IP and system design

Large FPGAs use multiple dies and high-bandwidth memory in one package.

Expected in your paper:

- Explain why FPGAs use multi-die packaging and HBM
- Describe architectures: silicon interposers and die crossings
- Discuss design impact: partitioning, timing, memory bandwidth
- Compare with chiplet trends in CPUs and GPUs

Starting points: SLR, HBM, interposer, chiplets, UCIe

FPGA Security: IP Protection and Post-Quantum Crypto

Topic 24 · IP and system design

FPGA designs face threats from IP theft to hardware Trojans; new cryptography is arriving.

Expected in your paper:

- Describe threats: IP theft, bitstream attacks, hardware Trojans, side channels
- Explain protection: bitstream encryption, IP encryption, PUFs
- Survey post-quantum cryptography accelerators on FPGAs
- Discuss supply-chain aspects of third-party IP

Starting points: bitstream encryption, PUF, hardware Trojan, PQC, Kyber

FPGA SmartNICs and In-Network Computing

Topic 25 · Applications and systems

FPGAs process network traffic in hardware close to the wire.

Expected in your paper:

- Explain SmartNIC architectures (FPGA, SoC, ASIC)
- Describe offload examples: packet processing, load balancing, encryption
- Compare programmability: P4, HLS, RTL
- Discuss uses in finance and cloud data centers

Starting points: SmartNIC, P4, DPDK, in-network computing, offload

FPGA Acceleration in the Cloud

Topic 26 · Applications and systems

Cloud providers offer FPGAs as accelerators that customers can program.

Expected in your paper:

- Survey cloud FPGA offerings and their shells
- Describe the programming and deployment flow
- Analyze workloads that benefit: search, compression, genomics, finance
- Compare cost and performance with GPUs

Starting points: AWS F2, Alveo, Vitis, XRT, accelerator-as-a-service

FPGAs for Software-Defined Radio and 5G/6G

Topic 27 · Applications and systems

Radio signal processing is a classical FPGA domain and is moving toward RF-integrated devices.

Expected in your paper:

- Explain the SDR signal chain and why FPGAs fit
- Describe RF-integrated devices (RFSoC) and their use in radio units
- Discuss DSP IP: DDS, FIR, FFT and massive MIMO processing
- Evaluate trends toward AI-native radio

Starting points: RFSoC, O-RAN, massive MIMO, DDC/DUC, JESD204

FPGAs in Automotive Systems and Functional Safety

Topic 28 · Applications and systems

Automotive systems use FPGAs for sensor processing under strict safety requirements.

Expected in your paper:

- Describe use cases: ADAS sensor fusion, domain controllers
- Explain ISO 26262 concepts: ASIL, safe design, tool qualification
- Discuss safety mechanisms: lockstep, ECC, redundancy
- Present certified IP and devices

Starting points: ISO 26262, ASIL, lockstep, ADAS, tool qualification

FPGAs in Space: Radiation Tolerance

Topic 29 · Applications and systems

Space missions use FPGAs for on-board processing and need protection against radiation.

Expected in your paper:

- Explain radiation effects (TID, SEE) and mitigation
- Compare radiation-tolerant, radiation-hardened and commercial FPGAs
- Describe on-board processing examples such as imaging and AI
- Discuss reconfiguration in orbit

Starting points: SEU, TID, rad-hard, CubeSat, in-orbit reconfiguration

FPGAs in Quantum Computing Control Systems

Topic 30 · Applications and systems

Quantum computers need fast, precise, many-channel control electronics.

Expected in your paper:

- Explain the control and readout needs of quantum computers
- Show why FPGAs and RFSoCs are used: latency, channel count, timing
- Describe open-source control stacks and example systems
- Discuss scalability challenges

Starting points: QICK, RFSoC, qubit control, feedback, timing

Next steps

Now

Form your team and read the topic slides carefully.

By 29.10.2026

Make your topic selection via moodle.
Only one topic per team
A topic cannot be selected by multiple teams
First come first serve

To make sure you get what you want: Prepar three preferred topics

Then

Start the literature research and prepare the outline.

Thank you for your attention!

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